TECHNICAL ARCHITECTURE PROPOSAL

AI inference deployment — fictional workshop scenario

1. Context and scope

The engineering team is planning a pilot for an internal document assistant. The pilot will serve employees in the European Union. The team must select how to run inference before implementation.

2. Non-functional requirements

The three highest-priority requirements for this proposal are:

1. Availability: at least 99.9% measured monthly.
2. Latency: 95th percentile response time below 3 seconds.
3. Data residency: process and store pilot data in the EU.

3. Alternatives under consideration

A. Hosted inference: use a managed provider's inference API.

Benefits: faster rollout and less infrastructure maintenance.
Limitations: provider dependency and variable usage charges.

B. Self-hosted inference: operate models on dedicated GPUs.

Benefits: greater control over infrastructure and deployment.
Limitations: GPU capacity and specialist operations are needed.

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3. Alternatives under consideration (continued)

C. Hybrid inference: combine hosted and self-hosted services.

Benefit: flexibility to route workloads to either environment.

Limitation: managing two environments increases complexity.

4. Recommendation

Use hosted inference for the pilot because it accelerates rollout and reduces the infrastructure work required by the small team.

Select an EU processing region and verify the residency contract.

Review self-hosting later if workload or control needs change.

5. Risks and estimated costs

Risks include provider dependency, unpredictable usage costs, and meeting the latency target during peak demand. Mitigate them with monitoring, budget alerts and a tested exit strategy.

Planning estimate for hosted inference: EUR 2,000 per month.

Planning estimate for self-hosted inference: EUR 5,000 per month.

These are estimates for planning, not actual production costs.